Dynamic network analysis of multi-scale -omics data for protein function prediction

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Introduction

- Understanding protein function is important
- Determining protein function experimentally is challenging
  - Traditional wet lab experiments for determining protein function are low-throughput
  - Only ~0.1% of protein sequences are reliably annotated
- Computational approaches for protein function prediction are needed
  - Sequence-based approaches
  - 3D structure-based approaches
  - Protein-protein interaction (PPI) network-based approaches

The UniProt Consortium, "UniProt: the Universal Protein Knowledgebase in 2023", Nucleic Acids Research, 2023

Introduction

Sequence

TTCCPSIVARS
NFNVCRLPGT
PEAICATYTGC
IIIPGATCPGD
YAN…

Function

- Enzyme
- Hormone
- Transport
- Storage
- …

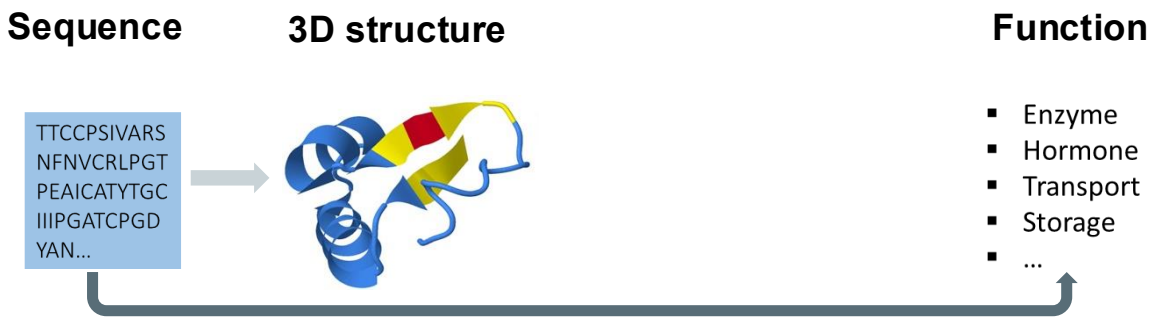
Computational approaches:

- Sequence-based
  - Pro: Lots of sequence data is available
  - Con: Dissimilar sequences may share functions Why?

Both are inhibitors of serine proteases

Gao, Z., Jiang, C., Zhang, J. et al. Hierarchical graph learning for protein–protein interaction. Nat Commun 14, 1093 (2023). https://doi.org/10.1038/s41467-023-36736-1

Introduction



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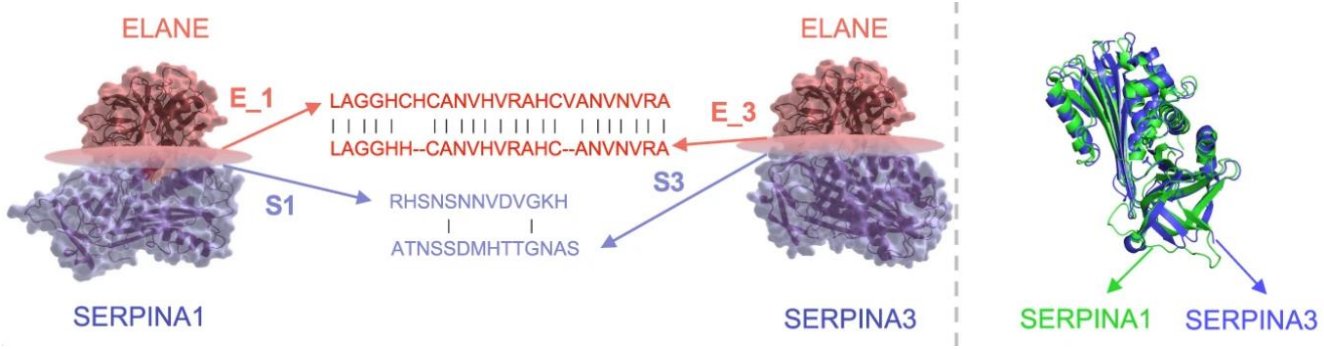

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A protein's 3D structure directs what other proteins it interacts with, and thus what function(s) it performs

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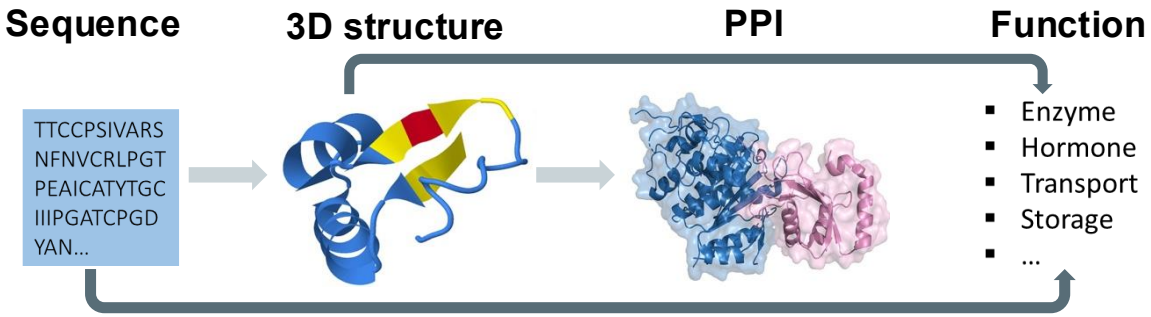

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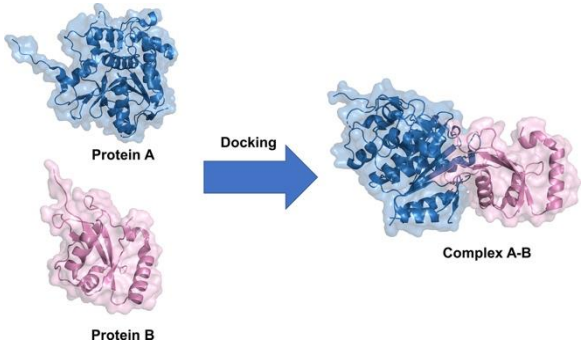
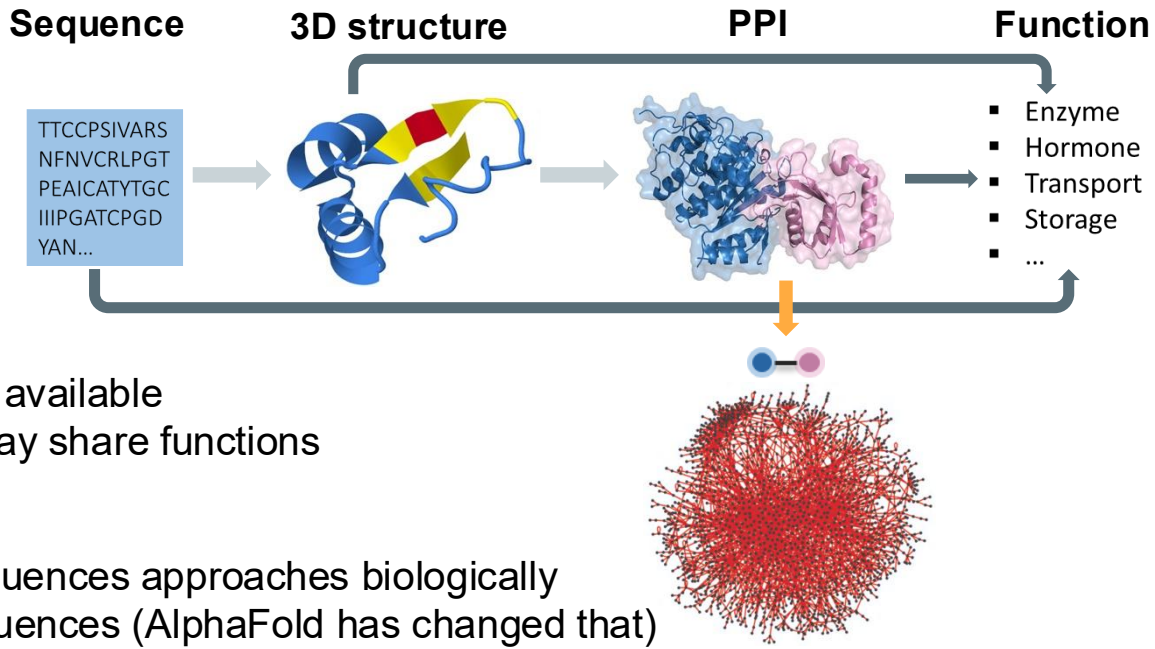
  - Pro: More promising than sequences approaches biologically
  - Con: Less data than with sequences (AlphaFold has changed that)



Proteins interact with other proteins through complexes to carry out their functions

Stanzione et al., "Use of molecular docking computational tools in drug discovery", Progress in medicinal chemistry, 2021

Introduction



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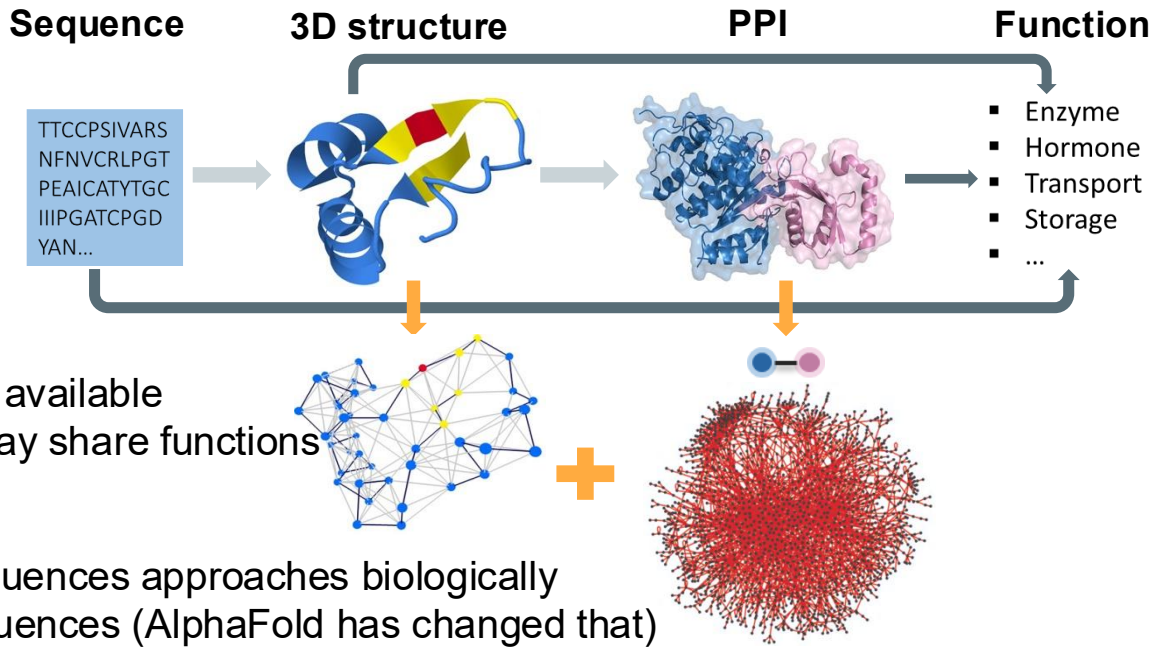
- Protein-protein interaction (PPI) network-based
  - Nodes are proteins; edges are PPIs
  - Pro: Physical bindings, i.e., protein complexes, are efficient in protein function prediction
  - Con: PPI data are incomplete and noisy

Chen et al., "Construction and analysis of protein-protein interaction network of heroin use disorder", Scientific reports, 2019

Introduction



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  - Pro: Lots of sequence data is available
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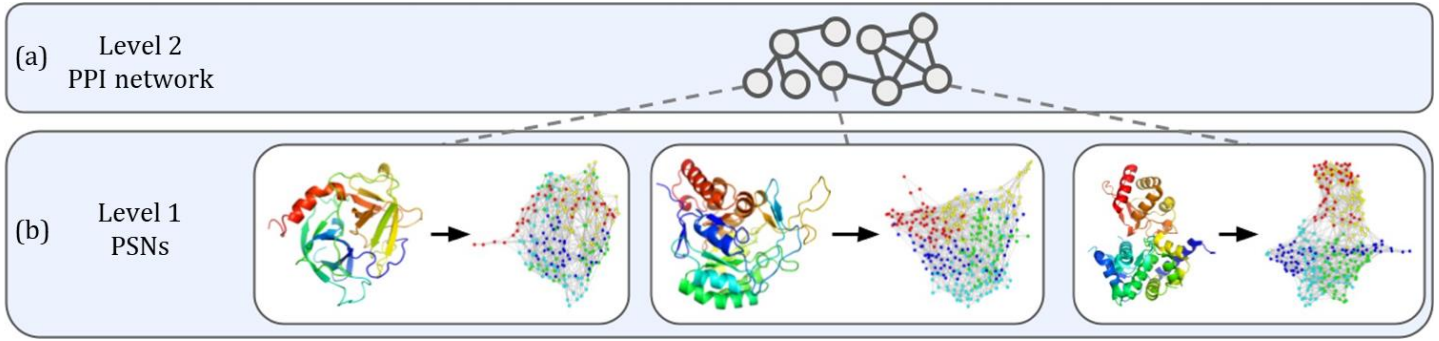
- 3D structure-based
  - Pro: More promising than sequences approaches biologically
  - Con: Less data than with sequences (AlphaFold has changed that)
  - 3D structure → protein structure network (PSN)
    - Nodes are amino acids; edges join amino acids close in 3D space
- Protein-protein interaction (PPI) network-based
  - Nodes are proteins; edges are PPIs
  - Pro: Physical bindings, i.e., protein complexes, are efficient in protein function prediction
  - Con: PPI data are incomplete and noisy

Chen et al., "Construction and analysis of protein-protein interaction network of heroin use disorder", Scientific reports, 2019

Introduction

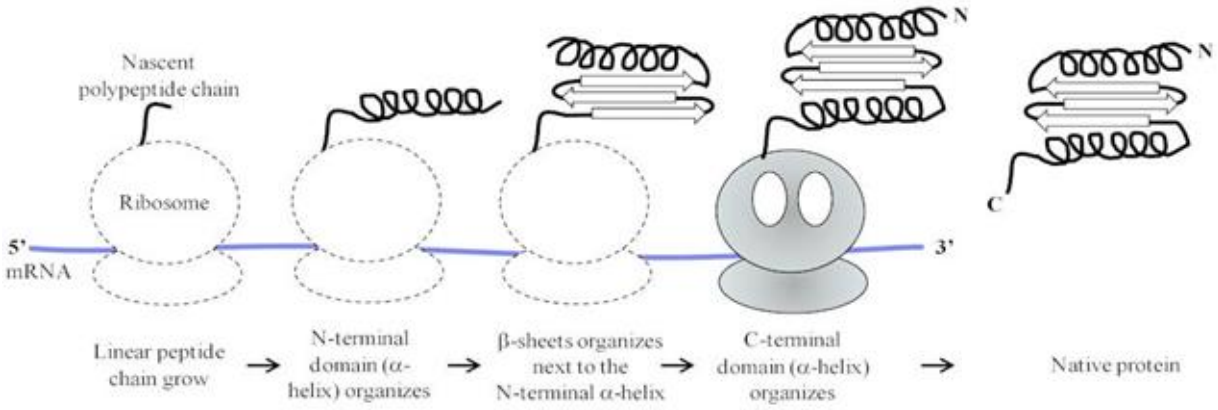
- Network of networks (NoN) approach
  - Recently, our lab integrated PSN and PPI data into a multi-scale NoN
    - At the PPI network level, each protein is a node in the PPI network
    - At the lower level, each protein is a PSN
  - Pro: For 30% of the GO terms, only NoN approaches made non-random predictions
  - Con: NoN approaches outperformed PSN-alone and PPI network-alone approaches in "only" ~50% of the GO terms



Gu et al., "Modeling multi-scale data via a network of networks", Bioinformatics, 2022

Introduction

- Until recently, final (native) 3D structures were modeled as static PSNs
- However, protein folding is dynamic, i.e., certain protein parts fold before others

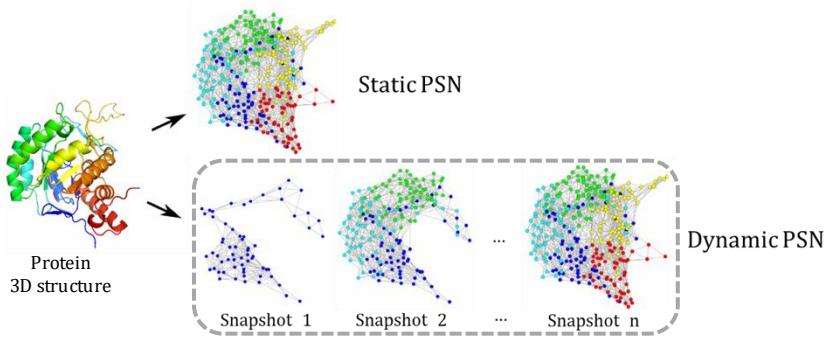


Biyani et al., "Solid-phase cell-free protein synthesis to improve protein foldability", Cell-Free Protein Synthesis, 2012

Introduction

- Until recently, final (native) 3D structures were modeled as static PSNs
- However, protein folding is dynamic, i.e., certain protein parts fold before others
- Our lab recently introduced dynamic PSNs
  - A dynamic PSN consists of multiple PSN snapshots, as visualized
  - Significantly outperformed static PSNs in the tasks of:
    - Protein structure classification: *Newaz, K. et al., Proteins: Structure, Function, and Bioinformatics, 2022*
    - Predicting protein structural change upon ligand binding: *Wells, A., Siyu, Y. et al., in preparation, 2024*

3D-SIG talk at 17:40 pm on 07/15

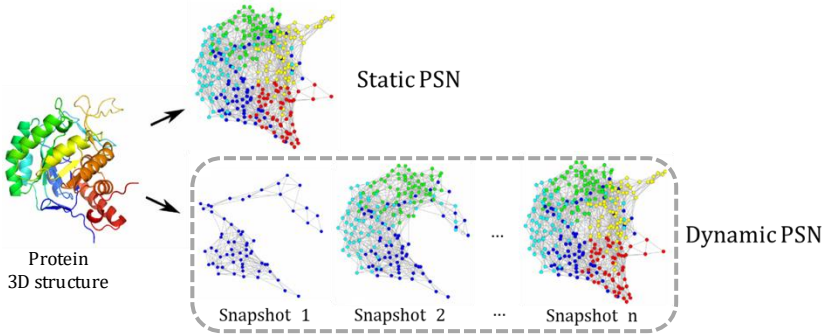


Newaz et al., "Multi-layer sequential network analysis improves protein 3D structural classification", Proteins: Structure, Function, and Bioinformatics, 2022

Our contribution

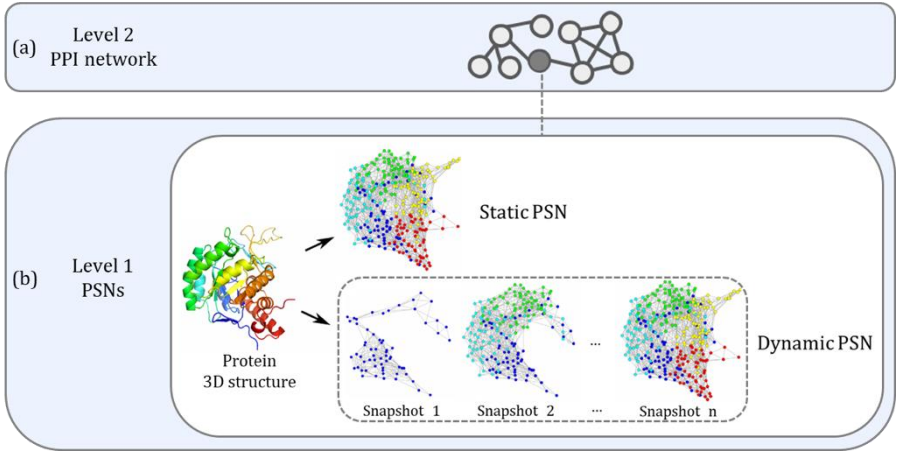
Will modeling protein 3D structures as dynamic PSNs benefit protein function prediction?

When analyzing PSNs only



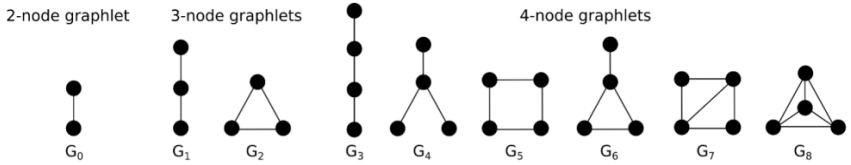
Previous work: dynamic PSNs in the other two tasks
vs.
Our work: dynamic PSNs in protein function prediction

When analyzing PSNs within the NoN framework

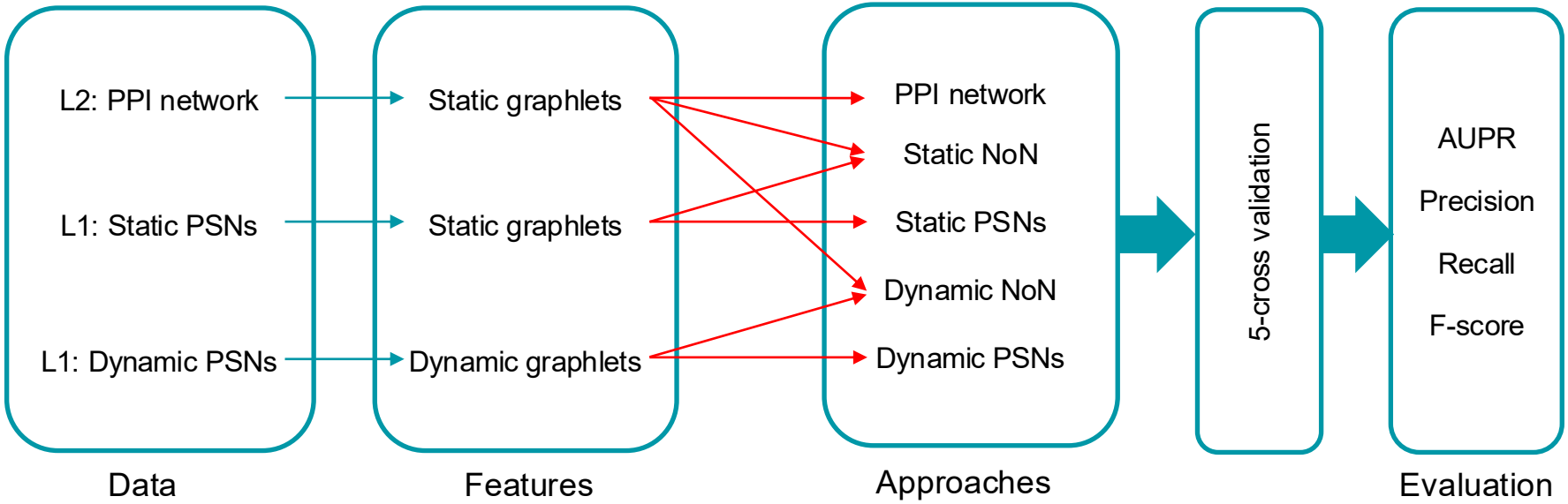


Previous work: An NoN with static PSNs in protein function prediction
vs.
Our work: An NoN with dynamic PSNs in protein function prediction

Data and methods

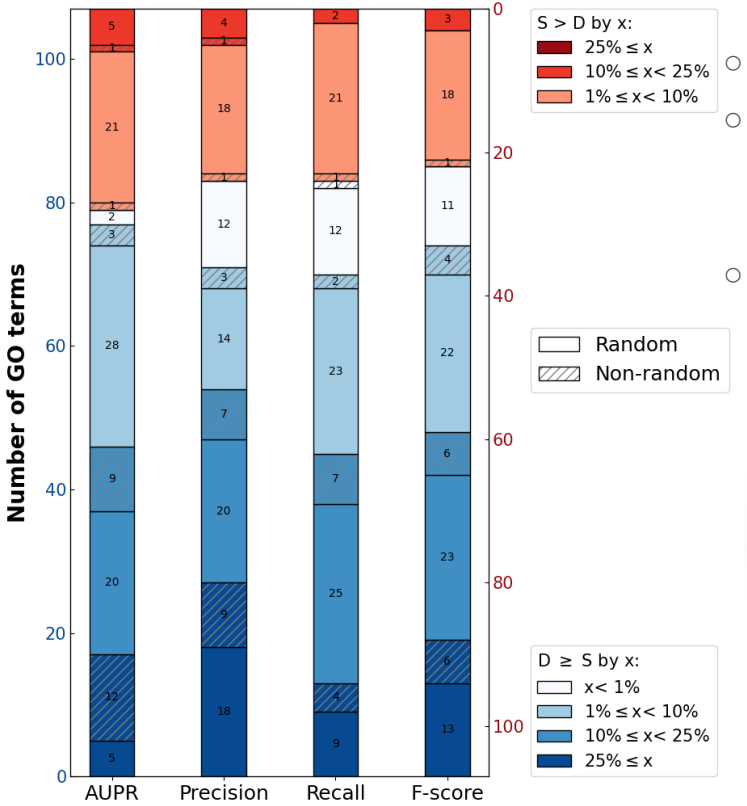


- PPI network: Human PPI network from BioGRID (18,708 nodes, 434,527 edges)
- PSNs: All 4,364 proteins from the PPI network with 3D structures in the Protein Data Bank (PDB)
- Labels for binary classification: All 107 GO terms that annotate 20+ of the 4,364 proteins

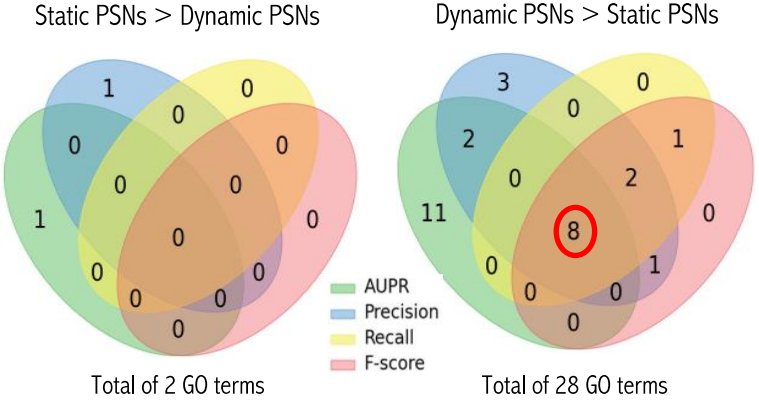


Results

- Static vs. dynamic PSNs when analyzing PSNs only



- Results for most GO terms are random for both approaches
- Of the non-random results (for at least one measure):
  - Static PSNs > Dynamic PSNs: 3 GO terms
  - Dynamic PSNs > Static PSNs: 30 GO terms
- When a given approach is non-random and superior to the other approach by at least 10%:

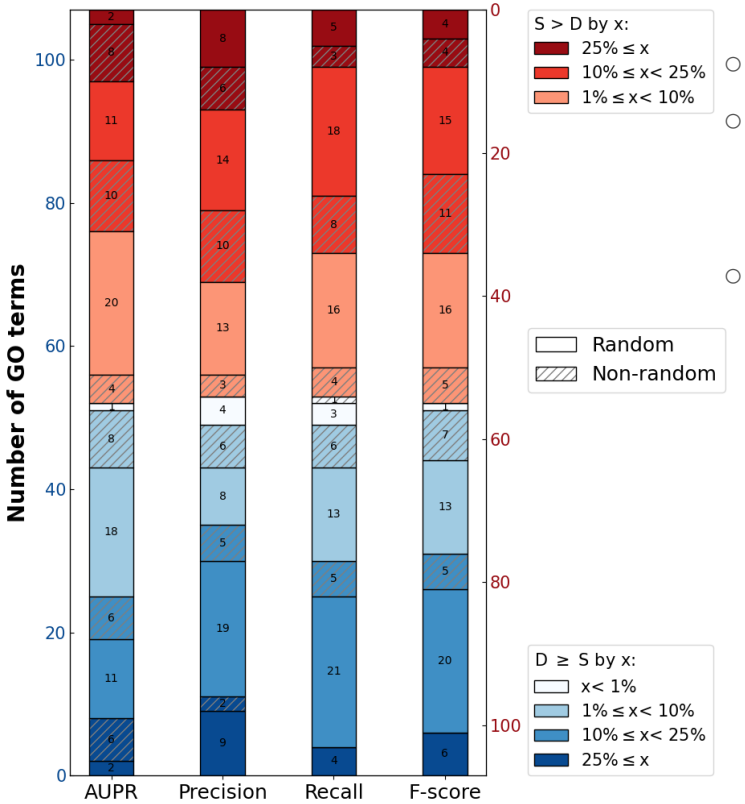


The 8 GO terms:
GO:0006470: protein dephosphorylation
GO:0007160: cell-matrix adhesion
GO:0017148: negative regulation of translation
GO:0018105: peptidyl-serine phosphorylation
GO:0018108: peptidyl-tyrosine phosphorylation
GO:0046777: protein autophosphorylation
GO:0051607: defense response to virus
GO:0070936: protein K48-linked ubiquitination

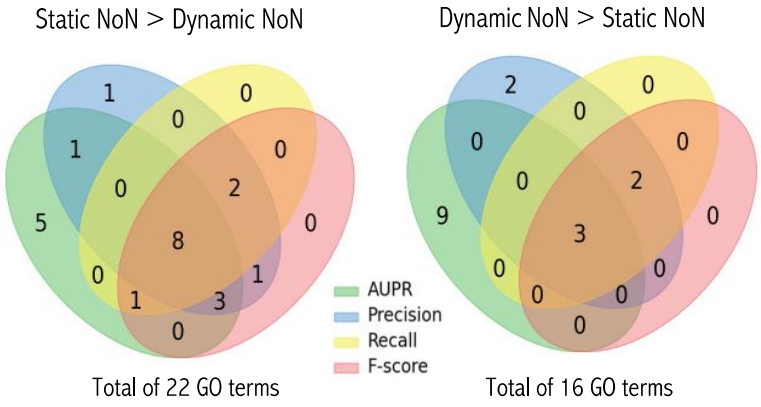
4/8 = 50% here
vs.
10/107 = 9% in the entire dataset

Results

- When analyzing static vs. dynamic PSNs within the NoN framework



- Still many random results for both approaches
- Of the non-random results (for at least one measure):
  - Static NoN > Dynamic NoN: 27 GO terms
  - Dynamic NoN > Static NoN: 24 GO terms
- When a given approach is non-random and superior to the other approach by at least 10%:



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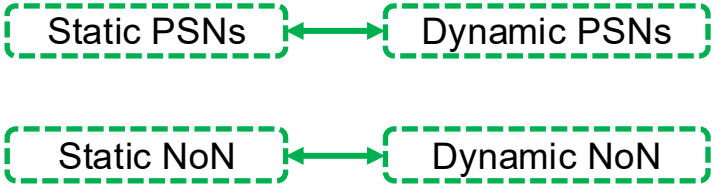


- When analyzing PSNs only
- When analyzing PSNs within the NoN framework

Results

Static PSNs

Static NoN

vs.

Dynamic PSNs

Dynamic NoN

- When analyzing PSNs only
- When analyzing PSNs within the NoN framework
- When analyzing all static vs. all dynamic approaches

Results

Static PSNs
OR
Static NoN
>
Dynamic PSNs
AND
Dynamic NoN

- When analyzing PSNs only
- When analyzing PSNs within the NoN framework
- When analyzing all static vs. all dynamic approaches
  - At least one static approach is non-random and better than both dynamic approaches
    - Result for at least one measure: 27 GO terms

Results

Static PSNs
AND
Static NoN

<

Dynamic PSNs
OR
Dynamic NoN

- When analyzing PSNs only
- When analyzing PSNs within the NoN framework
- When analyzing all static vs. all dynamic approaches
  - At least one static approach is non-random and better than both dynamic approaches
    - Result for at least one measure: 27 GO terms
  - At least one dynamic approach is non-random and better than both static approaches
    - Result for at least one measure: 27 GO terms

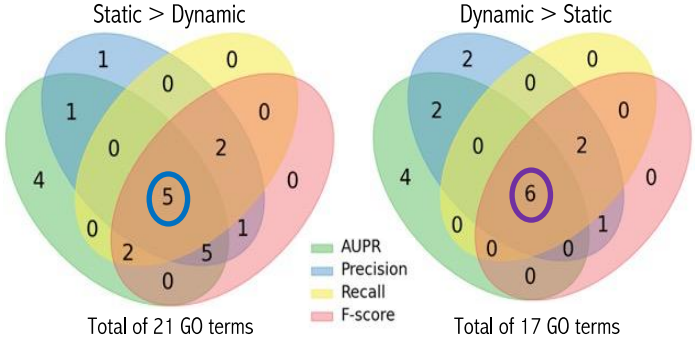
Results

Static PSNs
PPI network
Static NoN

vs.

Dynamic PSNs
Dynamic NoN

- When analyzing PSNs only
- When analyzing PSNs within the NoN framework
- When analyzing all static vs. all dynamic approaches
  - At least one static approach is non-random and better than both dynamic approaches
    - Result for at least one measure: 27 GO terms
    - It is always the static NoN that is superior → importance of considering NoN!
  - At least one dynamic approach is non-random and better than both static approaches
    - Result for at least one measure: 27 GO terms
    - It is either the dynamic NoN or dynamic PSNs (or both) that are superior
  - When we set the superiority threshold to at least 10%:



Results

Even when considering the PPI network approach, all of dynamic PSNs, dynamic NoN, and static NoN continue to have value

Static PSNs		Dynamic PSNs
PPI network	**vs.**	
Static NoN		Dynamic NoN

- When analyzing PSNs only
- When analyzing PSNs within the NoN framework
- When analyzing all static vs. all dynamic approaches
  - At least one static approach is non-random and better than both dynamic approaches
    - Result for at least one measure: 27 GO terms → 49 GO terms
    - It is always the static NoN that is superior → importance of considering NoN!
  - At least one dynamic approach is non-random and better than ~~both~~ all three static approaches
    - Result for at least one measure: 27 GO terms → 26 GO terms
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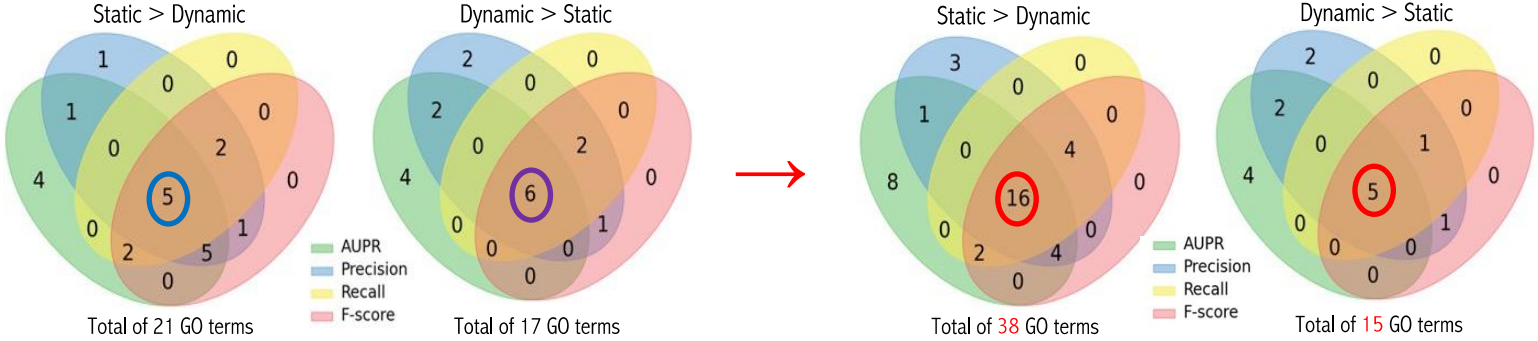


Conclusions and future directions

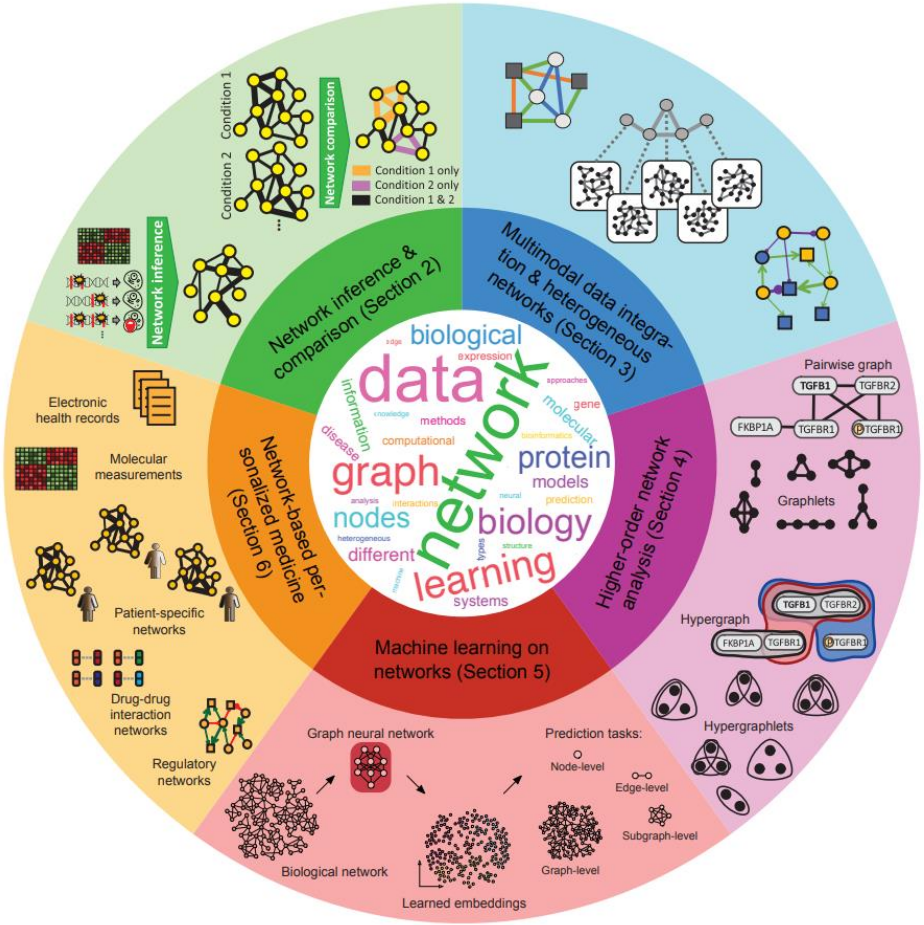
- In all analyses, dynamic PSNs bring some value
  - Dynamic PSNs still made random predictions for most GO terms (so do all other approaches!)
- The NoN framework (static as well as dynamic) also brings value
- No approach works well universally. E.g., for AUPR:
  - For 40 GO terms (37%), all five approaches yield random results
  - Of non-random results, even the best approach (PPI network) wins for only 30 GO terms (28%)
- Why do PSNs, i.e., 3D structures, provide less predictive power than PPIs?
- We cannot claim statistical significance for many results after corrections

Conclusions and future directions

- New methodological improvements needed?
  - Different data, tasks, and applications
  - Code/data typically not provided

Existing NoN studies	Data type		Network analysis task	Application
Ni et al., BMC Bioinformatics, 2016.	Level 2: disease similarity network	Level 1: tissue-specific PPI networks	Node ranking	Candidate disease gene prioritization
Wang et al., IJCAI, 2021. Wang et al., WWW, 2022.	Level 2: Network of interacting molecules	Level 1: molecular graphs	Link prediction	Prediction of interactions between molecules (e.g., drugs)
Chu et al., Information Sciences, 2022.	Level 2: Network of interactions between drugs and their target proteins	Level 1: drug molecule graphs and the targets' PSNs	Link prediction	Prediction of interactions between drugs and their target proteins
Gao et al., Nature Communications, 2023.	Level 2: PPI network	Level 1: static PSNs	Link prediction	Prediction of interactions between proteins
Gu et al., Bioinformatics, 2022 (Our previous work)	Level 2: PPI network	Level 1: static PSNs	Entity label prediction (node/network classification)	Protein function prediction
This work	Level 2: PPI network	Level 1: dynamic PSNs	Entity label prediction (node/network classification)	Protein function prediction

Our lab's perspective paper: Current and future directions in network biology, *Bioinformatics Advances*, 2024.



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ntx nature's toolbox

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